the laws of quantum mechanics and nuclear physics. During the preceding twenty years, he had developed powerful mathematical models to describe how atomic nuclei behave; with Niels Bohr he had developed the laws of nuclear fission (the splitting apart of heavy atomic nuclei such as uranium and plutonium, the principle underlying the atomic bomb); and he had been the leader of a team that designed the American hydrogen bomb (Chapter 6). Drawing on this expertise, Wheeler guided Harrison through the intricacies of the analysis.

The result of their analysis, the equation of state for cold, dead matter, is depicted and discussed in Box 5.5. At the densities of white dwarfs, it was the same equation of state as Chandrasekhar had used in his white-dwarf studies (Chapter 4); at neutron-star densities, it was the same as Oppenheimer and Volkoff had used (Box 5.4); at densities below white dwarfs and between white dwarfs and neutron stars, it was completely new.

With this equation of state for cold, dead matter in hand, John Wheeler asked Masami Wakano, a postdoc from Japan, to do with it what Volkoff had done for neutron stars and Chandrasekhar for white dwarfs: Combine the equation of state with the general relativistic equation describing the balance of gravity and pressure inside a star, and from that combination deduce a differential equation describing the star's structure; then solve the differential equation numerically. The numerical calculations would produce the details of the internal structures of all cold, dead stars and, most important, the stars' masses.

The calculations for the structure of a single star (the distribution of density, pressure, and gravity inside the star) had required Chandrasekhar and Volkoff many days of effort as they punched the buttons of their calculators in Cambridge and Berkeley in the 1930s. Wakano in Princeton in the 1950s, by contrast, had at his disposal one of the world's first digital computers, the MANIAC—a room full of vacuum tubes and wires that had been constructed at the Princeton Institute for Advanced Study for use in the design of the hydrogen bomb. With the MANIAC, Wakano could crunch out the structure of each star in less than an hour.